

INTELLIGENT CLINICAL SYSTEMS

AI Medical Assistant

Intelligent Healthcare Assistant using local Generative AI & LLMs with Ollama framework, Phi-3 Mini, Python & Streamlit interface.

SUBMITTED BY

Rahul Singh

TECH STACK

Ollama, Phi-3, Streamlit

ACADEMIC YEAR

2025 – 2026

Executive Abstract

Developing an innovative healthcare-based application designed to provide intelligent and automated medical guidance.

-  **Symptom Analytics:** Generates automated suggestions using local Large Language Models (LLMs).
-  **Absolute Privacy:** Completely local AI processing through **Ollama** framework. No third-party API dependencies.
-  **Direct Documentation:** Automated preliminary diagnostic reports formatted in PDF natively via Python.



| System Introduction



Healthcare Evolution

AI-powered assistants are transforming multiple domains. In healthcare, virtual systems act as preliminary checkpoints, enhancing medical accessibility and dramatically reducing first-contact overhead.



Core Project Motivation

Many individuals struggle to gauge symptom severity, causing clinical overload or dangerous delays. This virtual medical assistant bridges this gap, providing clinical guidance directly at home.

Problem Statement



Delayed Care

Physical healthcare scheduling and long facility wait times delay preliminary diagnosis for critical health concerns.



Privacy Gaps

Cloud-based clinical models pose privacy threats to patient data. Offline tools are essential to shield health confidentiality.



Poor Accessibility

Remote or rural sectors suffer from limited physical medical consulting resources and immediate diagnostic setups.

The Proposed Solution

Leveraging Edge Intelligence for Decentralized Healthcare

- 🛡️ **Offline AI Inference:** Powered completely by locally-hosted LLMs using Ollama framework.
- 🏠 **Proactive Triage:** Immediate identification of severe conditions with integrated emergency triggers.
- 📋 **Specialist Routing:** Intelligent guidance recommending correct specialists based on specific symptoms.



| Core Objectives

- ✓ **Natural Language Comprehension:**

Process unformatted symptoms in plain text input from patients like a normal conversation.

- ✓ **Automated Diagnostics Engine:**

Suggest probable illnesses, severity assessments, precautions, and actionable clinical advice.

- ✓ **PDF Report Architecture:**

Programmatic generation of a standardized patient summary in PDF via Python FPDF engine.



Crucial Disclaimer

This platform serves strictly as a preliminary triage aid and informational guide. It does NOT substitute professional medical consultation or final hospital diagnostic workflows.

System Requirements

Category	Minimum Requirements	Recommended (Optimal)
Processor (CPU)	Intel Core i3 or AMD equivalent (64-bit)	Intel Core i7 / AMD Ryzen 7 or higher
System Memory (RAM)	8 GB RAM	16 GB RAM (Highly beneficial for LLM context speed)
Frameworks & LLMs	Python 3.10+, Streamlit, Ollama Runtime	Phi-3 Mini (3.8B parameters) local installation
Python Libraries	fpdf, ollama, streamlit, datetime	Active libraries verified on standard Pip package registry
OS Environment	Windows 10 / Ubuntu 20.04 LTS / macOS	Windows 11 / Unix environment with GPU drivers (CUDA)

System Architecture



1. Client Interface

The patient inputs symptoms, name, age, and sex via a responsive, custom styled Streamlit web interface.



2. Local AI Pipeline

Streamlit directs data to local Ollama. Phi-3 mini executes high-fidelity inference locally utilizing tailored prompting schemas.



3. Output & PDF

The system validates emergency keywords, generates specialist matches, and compiles instant downloadable PDF reports via FPDF.

Major Modules



AI Diagnostics Module

Utilizes custom system instructions to force the Phi-3 Model to systematically break symptoms into structured blocks: probable causes, exact severity levels, standard clinical warnings, and safe recovery steps.



Dynamic FPDF Report Engine

Constructs dynamic document structures to prevent text overflows. Captures clinical output, maps patient identity variables, adds precise timestamps, and processes safe rendering onto a standard medical template layout.

| Emergency Detection

100%

REAL-TIME ALERTS

Proactive Risk Assessment

Safety is paramount. Before invoking the LLM model, the Python core backend operates a regex keyword filter analyzing input against highly critical conditions.

- ! **Critical Triggers:** Chest pain, breathing difficulty, stroke symptoms, loss of consciousness.
- ! **Automated Triage UI:** Instantly displays high-risk red indicators recommending emergency hotlines.

| Specialist Routing

Intelligent routing pipeline matches basic clinical keywords with appropriate doctors to optimize patient care pathing:



Cardiologist

Assigned to: Chest tightness, palpitations, heavy breathing.



Dermatologist

Assigned to: Severe skin rashes, sudden lesions, scaling.



Neurologist

Assigned to: Numbness, persistent severe migraines, vertigo.



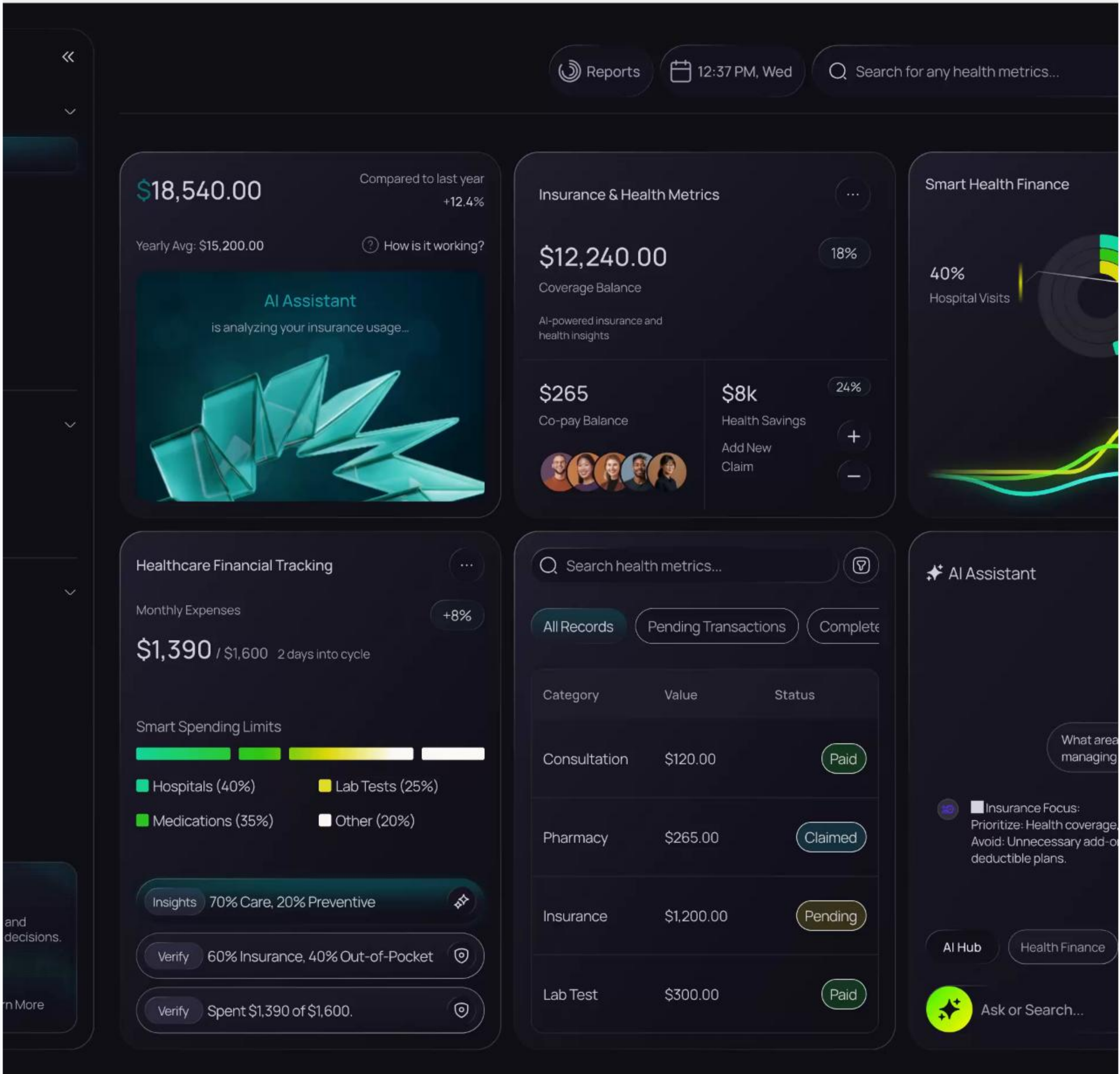
General Physician

Assigned to: Seasonal fevers, typical flu symptoms, common cold.

User Interface

Streamlit dashboard customized with clean medical styling components:

- ✓ **Simple Forms:** Direct input interface capturing personal variables.
- ✓ **Formatted Cards:** AI medical responses rendered with clear visual segmentation.
- ✓ **Action Center:** Instant report compile action and direct downloadable triggers.



Testing & Validation

Test ID	Input Case	Expected Clinical Outcome	Result
TC01	Fever, body fatigue, chills	Viral infection suggestion, hydration guidance, GP consult	✓ PASS
TC02	Severe chest pain	Emergency trigger matches chest keywords, prompts Red Alert warning	✓ PASS
TC03	Itchy hives, red skin lesions	Assesses dermatological issues and recommends Dermatologist routing	✓ PASS
TC04	No text / Blank input space	Triggers robust standard validator asking user to describe symptoms	✓ PASS

Evaluation Matrix

Key Advantages

- ✓ High speed locally-hosted model execution
- ✓ 100% data protection & offline security
- ✓ Elimination of subscription or cloud API overhead

Limitations

- ✗ Inability to provide absolute clinical diagnosis
- ✗ Inference speeds are hardware configuration dependent
- ✗ General knowledge limits on minor custom anomalies

Future & Conclusion

A critical step forward in integrating locally hosted, ultra-secure Large Language Models into daily diagnostic workflows.

Voice Diagnostics

Integrating speech-to-text workflows for accessible hands-free symptom checking.

Database Tracking

Incorporating SQL secure schemas to persistently track patients health logs over time.

Real-time IoT Sync

Connecting smartwatch interfaces to monitor temperature and pulse anomalies dynamically.

 rahul.singh@academic.edu

 localhost:8501 (Streamlit Entry)

Image Sources



https://www.it-united.at/wp-content/uploads/2025/03/KI-im-Gesundheitswesen-_AdobeStock_1285247887-1-1400x794.jpg

Source: www.it-united.at



https://plus.unsplash.com/premium_photo-1764695595385-ed92175c3857?fm=jpg&q=60&w=3000&auto=format&fit=crop&ixlib=rb-4.1.0&ixid=M3wxMjA3fDB8MHxwaG90by1wYWdlfHx8fGVufDB8fHx8fA%3D%3D

Source: unsplash.com



<https://cdn.dribbble.com/userupload/45213248/file/still-018800a46129830d9ea923953f53e424.png>

Source: dribbble.com